

1. a) To determine whether it is in BCNF
 i) or not, we need to check A, BC and DE are candidate key or not.

$A^+ \rightarrow AE$ As they are not candidate
 $(BC)^+ \rightarrow BCAB$ Key. So, the relation
 $(DE)^+ \rightarrow DEB$ R is not in BCNF.

ii) $R_1(A, E)$ Now,
 $R_2(A, B, C, D)$ $R_1 \cap R_2 = AE \cap ABCD$
 $= A$
 $A^+ \rightarrow AE = R_1$
 So, the decomposition is lossless.

$A \rightarrow E$ can be checked from R_1
 $BC \rightarrow A$ " " " " R_2 } So, these
 dependencies are preserving.

For, $DE \rightarrow B$

result = DE

$$\begin{aligned}
 t &= (\pi_{\text{result}} \cap R_1)^+ \cap R_1 \\
 &= (DE \cap AE)^+ \cap AE \\
 &= E^+ \cap AE = E \cap AE = E
 \end{aligned}$$

$$\text{result} = DE \cup E = DE$$

again,

$$\begin{aligned} t &= (DE \cap ABCD)^+ \cap ABCD \\ &= D^+ \cap ABCD \\ &= D \cap ABCD = D \end{aligned}$$

$$\text{result} = DE \cup D = DE$$

So, result is not changing. So,

$$DE \rightarrow BE, \text{ so,}$$

$DE \rightarrow B$ is not preserved.

So, R is lossless decomposition but not dependency preserving.

b)

$$AB \rightarrow CD$$

$$A \rightarrow B$$

$$B \rightarrow C$$

$$C \rightarrow E$$

$$BD \rightarrow A$$

$$BD \rightarrow C$$

So,

$$C.K. = \{A, BD\}$$

$$A^+ \rightarrow ABCDE \checkmark$$

$$B^+ \rightarrow BCE \times$$

$$C^+ \rightarrow CE \times$$

$$D^+ \rightarrow D \times$$

$$E^+ \rightarrow E \times$$

$$(BC)^+ \rightarrow BCE \times$$

$$(BD)^+ \rightarrow BDCAE \checkmark$$

$$(BE)^+ \rightarrow BEC \times$$

$$(CD)^+ \rightarrow CDE \times$$

$$(CE)^+ \rightarrow CE \times$$

$$(DE)^+ \rightarrow DE \times$$

$$(BCE)^+ \rightarrow BCE \times$$

$$(CDE)^+ \rightarrow CDE \times$$

ii) $AB \rightarrow CD$	<u>Step-1:</u>	<u>Step: 2</u>
$A \rightarrow B$	$AB \rightarrow C$	$(AB)^+ \rightarrow ABCED$
$B \rightarrow C$	$AB \rightarrow D$	$A^+ \rightarrow ABCDE$
$C \rightarrow E$	$A \rightarrow B$	$B^+ \rightarrow BCE$
$BD \rightarrow A$	$B \rightarrow C$	$C^+ \rightarrow CE$
$BD \rightarrow C$	$C \rightarrow E$	$(BD)^+ \rightarrow BDACE$
	$BD \rightarrow A$	
	$BD \rightarrow C$	

Now,

Removing $AB \rightarrow C$,

$(AB)^+ \rightarrow ABDCE$, Same, So $AB \rightarrow C$ is redundant.

Removing $AB \rightarrow D$,

$(AB)^+ \rightarrow ABCE$, Not same.

Removing $A \rightarrow B$,

$A^+ \rightarrow A$, Not same.

Removing $B \rightarrow C$,

$B^+ \rightarrow B$, Not same.

Removing $C \rightarrow E$,

$C^+ \rightarrow C$, Not same.

Removing $BD \rightarrow A$,

$(BD)^+ \rightarrow BDC E$, Not same.

Removing $BD \rightarrow C$,

$(BD)^+ \rightarrow BDACE$, Same, So, $BD \rightarrow C$ is redundant.

After removing redundant rules,

$AB \rightarrow D$

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow E$

$BD \rightarrow A$

Step-3:

$A^+ \rightarrow ABCE D$

So, B is extraneous.

So, $AB \rightarrow D$ becomes

$A \rightarrow D$

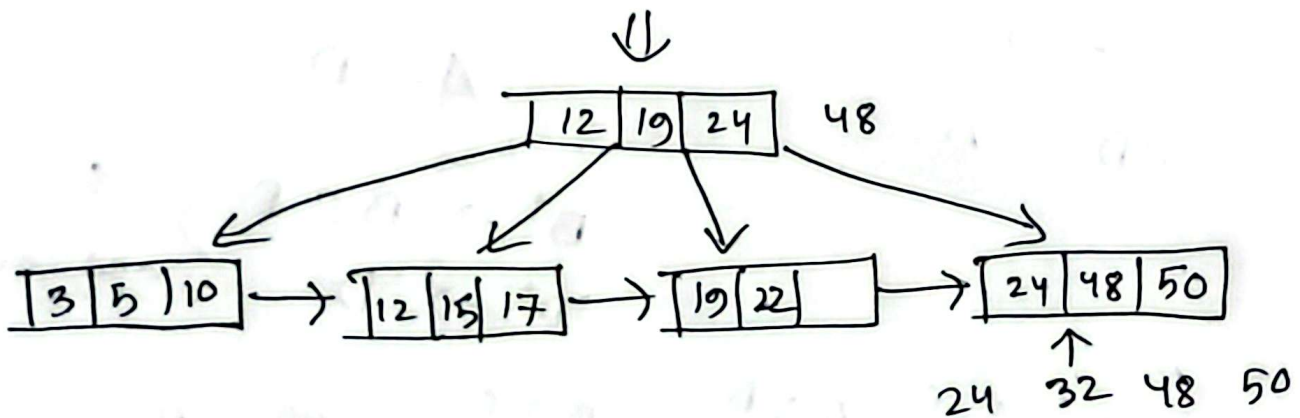
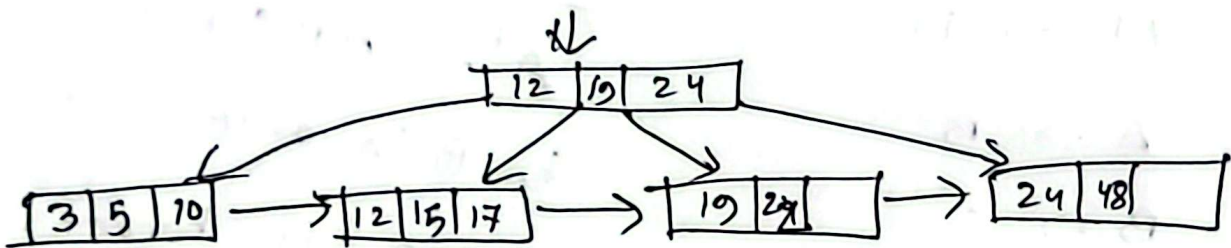
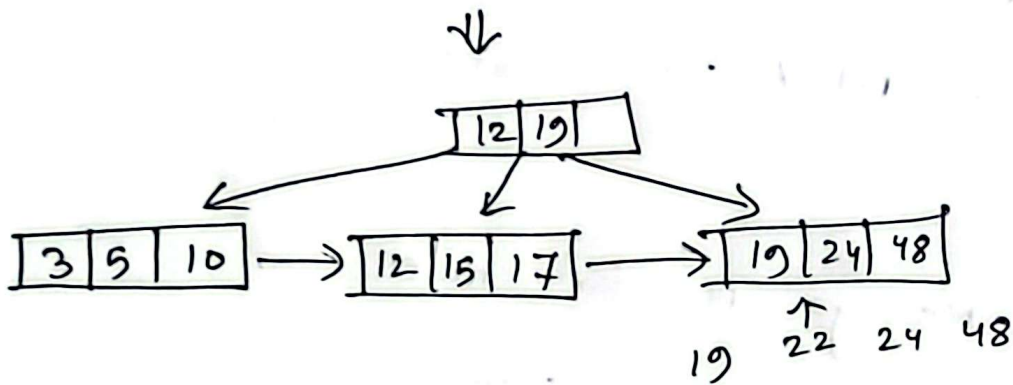
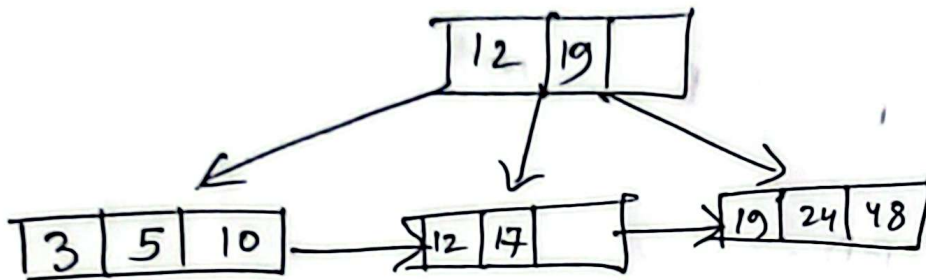
$B^+ \rightarrow BCE$ } So, none

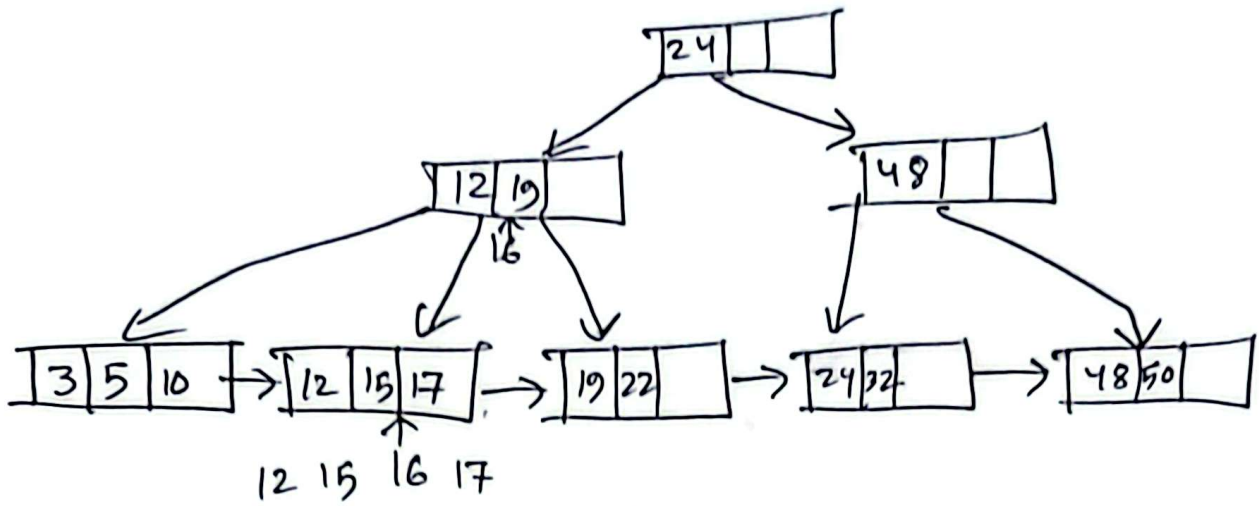
$D^+ \rightarrow D$ } are extraneous.

Minimal Cover: $\{ A \rightarrow D, B \rightarrow C, BD \rightarrow A, A \rightarrow B, C \rightarrow E \}$

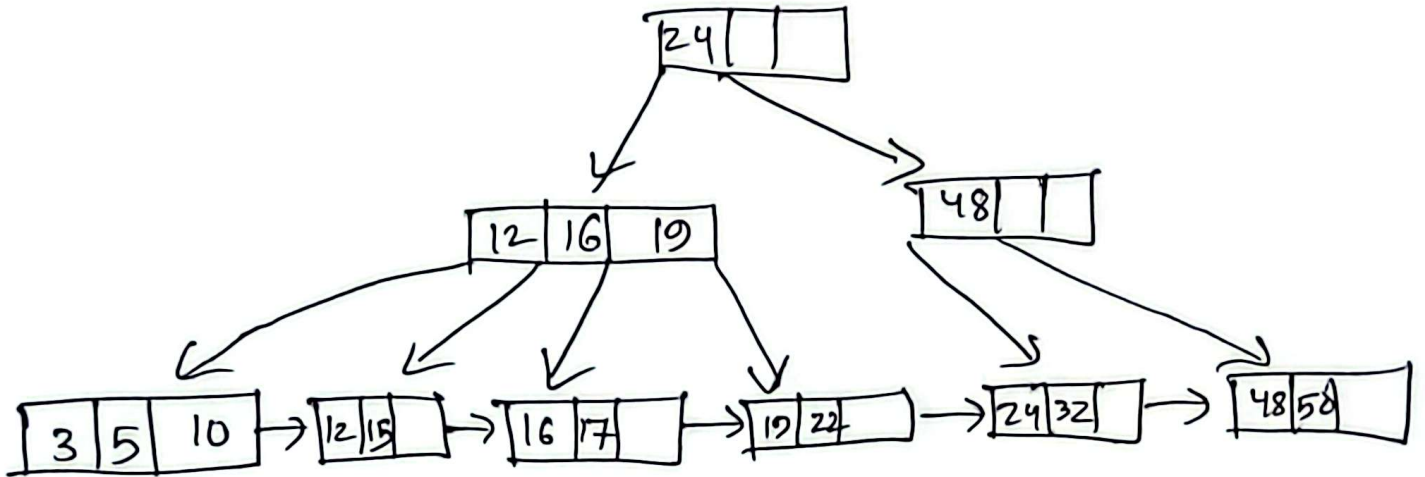
2. b) We can see that, the given B+ tree has max key = 3, So, order = $3 + 1 = 4$

10 15 22 50 32 16





↕



a) For $CD \rightarrow E$

To check its validity,

$$(CD)^+ \rightarrow CDF$$

So, we cannot get $CD \rightarrow E$. It is invalid.

For $AB \rightarrow CE$

$$A \rightarrow C$$

$$\Rightarrow AB \rightarrow BC ; [\text{Using Augmentation}]$$

$$\Rightarrow AB \rightarrow C ; [\text{Using Decomposition}]$$

Now, $AB \rightarrow C$ and $AB \rightarrow E$ (Given)

$$\text{So, } AB \Rightarrow CE ; [\text{Using union}]$$

b) i)

$$P \rightarrow SQR, Q \rightarrow T, R \rightarrow U$$

$$PR \rightarrow U, S \rightarrow Q$$

P is not present in R.H.S. So, P

P^+ must be in C.K.

$$P^+ \rightarrow PSQRUT$$

$$\text{So, C.K.} = \{P\}$$

ii) Step-1:

$$P \rightarrow S$$

$$P \rightarrow Q$$

$$P \rightarrow R$$

$$Q \rightarrow T$$

$$R \rightarrow U$$

$$PR \rightarrow U$$

$$S \rightarrow Q$$

Step-2 :

$P \rightarrow S$
 $S \rightarrow Q$ $\rightarrow$ so, $P \rightarrow Q$.
this rule is redundant.

$P \rightarrow R$
 $\Rightarrow PR \rightarrow R$ $\rightarrow$ so, $PR \rightarrow U$.
and $R \rightarrow U$ this rule is redundant.

So,

Minimal Cover =

$\{ P \rightarrow S, P \rightarrow R, Q \rightarrow T, \\ R \rightarrow U, S \rightarrow Q \}$

iii)

	<u>BCNF</u>	<u>3F</u>	<u>2F</u>
$P \rightarrow S$	✓		
$P \rightarrow R$	✓		
$Q \rightarrow T$	X	X	✓
$R \rightarrow U$	X	X	✓
$S \rightarrow Q$	X	X	✓

$$C.K. = \{P\}$$

$$P.A. = \{P\}$$

$$N.P.A. = \{Q, R, S, T, U\}$$

So, Highest normal form for

the relation = 2NF.

c) $L \rightarrow NR$

$$M \rightarrow OP$$

$$LM \rightarrow P$$

$$R1 \cap R2 = LM$$

$$(LM)^+ \rightarrow LMNROP = R1 \\ = R2$$

So, loss less decomposition.

$L \rightarrow NR$ will hold for $R1$

$M \rightarrow OP$ " " " $R2$

$LM \rightarrow P$ " " " $R2$.

So, FD will be preserved for the given decomposition.

4. b) order = 4 , so, max key = $4-1=3$

23 9 3 10 7 15 30 2 19 8 25 21 6 22 4

